

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

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AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one JSON action → execution evidence ↻

01

CONTROLLABLE SMALL LLMS

Choose the model boundary

- Execution speed
- Hardware and deployment
- Post-training access



02

ASKME LOOP

One small action at a time

```
plan → act → execute / test  
→ fresh evidence ↻
```

Interpret the result. Update locally when possible.



03

EXTERNAL ACCEPTANCE LAYER

Check the delivered workflow

- Preserve the full contract
- Check required behavior
- Accept the real artifact independently

The bridge: fixed action vocabulary + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

The retained record does not independently prove stdout or source contents.

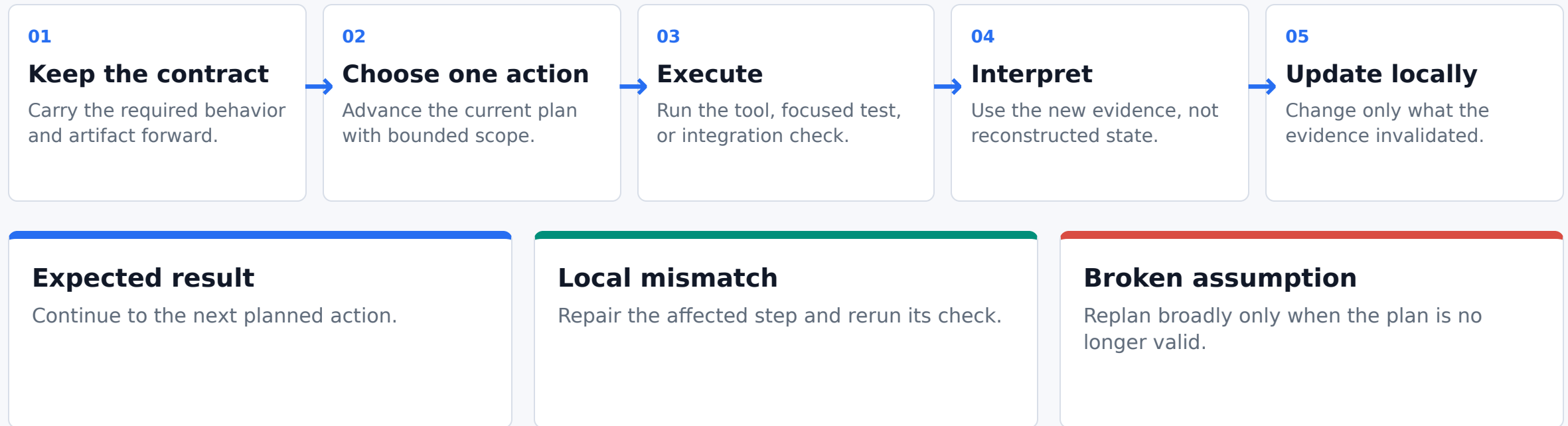
Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion $\neq$ accepted artifact.

Current limit: acceptance scored this run after completion; the failure was not returned for recovery.

Hypothesis: update only what fresh evidence invalidates



Trajectory goal—not yet isolated: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

Acceptance caught the one bad deliverable

Eight one-shot cells across four hosted variants

8 / 8

reported complete

7 / 8

artifacts accepted

The fastest build was the rejected one.
Completion and speed were not enough.

HOSTED VARIANT	ARTIFACT BUILD	SANITY REPAIR
Gemma 4 26B A4B	Accepted	Accepted
Gemma 4 31B	Accepted	Accepted
Qwen3.6-27B	Accepted	Accepted
Qwen3.6-35B-A3B	Wrong path	Accepted

Compatibility smoke, not a ranking: n=1/cell, descriptive only; no speed, family, size, reasoning, or reliability inference.

AskMe now makes FeatureBench edits

The same frozen task moved from no write to partially working code.

BEFORE · JULY CANARY

0 writes

Empty patch

The action interface blocked the edit.
The task stayed unresolved.

AFTER · AUG 1

Both patches applied

→ **11 / 13** **7 / 13**

Gemma target tests Qwen target tests
AskMe reached partially working code
in both attempts.

NEXT BOTTLENECK

No clean validation or finish

→

Gemma: 18 writes, 0 tests
Qwen: 1 write, then reading
Both: no done

What changed: AskMe now makes code changes; validation and termination are the remaining harness problems.

One task · one attempt/model · changed serving stack · evidence of progress, not a FeatureBench score or transport-only causal claim.

Promising for bounded loops. Feature readiness is still open.

Observed

Simple smoke: 7 / 8 artifacts accepted. On one feature task, revision 3 moved both models from empty to applied but unresolved patches: Gemma 11/13 target tests; Qwen 7/13.

Supported

Harness design was consequential on this task. The bundled changes moved failures into execution: Gemma rewrote without testing; Qwen resumed observation after one write.

Still open

Reliability beyond one task; clean validation and termination; transport versus serving/configuration effects; reasoning, family, size, architecture, and local performance. Revision 4 is implemented in open PR #21; v7 requalification is pending.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries—not a leaderboard.

	AskMe	pi	OpenHands
ACTION SURFACE	6 fixed JSON actions; exactly one action per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible <code>Action → Observation</code> tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	<code>done</code> → conditional fail-open validation; held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	<code>finish</code> signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.